Aabid Mohamed A

BACKEND & AI ENGINEER

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SUMMARY

Backend and AI engineer with production experience deploying **agentic systems on live banking infrastructure** at PwC — NLP extraction pipelines, ML classifiers, and async microservices handling ~50 deals/day. B.Tech in Electrical & Electronics Engineering, NIT Calicut; self-taught in systems and applied AI. Seeking backend and AI platform roles.

EXPERIENCE

PwC — Client: SMBC (Sumitomo Mitsui Banking Corporation) Aug 2025 – Present · Bengaluru

Associate Software Engineer

- Built an end-to-end agentic deal platform that eliminated a manual workflow for **25 salespeople** — async microservices processing **~50 deals/day** across ingestion, extraction, classification, and approval in under 60 seconds.
- Improved downstream ML extraction quality **30%+** with a chat-ingestion pipeline that normalized raw deal conversations into retrieval-ready structured documents.
- Eliminated manual entity tagging with a prompt-driven NLP layer (investor, tranche, deal size, terms) and per-call evaluation logging that catches extraction drift before scoring.
- Cut irrelevant blotter entries **~40%** with an ML deal-likelihood classifier, and delivered a blotter API plus live potlist that held **zero downtime** for 25 concurrent users and saved **~2 hours/day** in status checks.

SchoolHack.AI Jan 2025 – Feb 2025 · Remote

Machine Learning Intern

- Cut query latency below **3 seconds** by building a multi-source research agent that orchestrates Tavily web search and Hunter data retrieval into structured outputs.
- Reduced manual SDR prospecting **~60%** with a LinkedIn and Gmail outreach agent automating contact discovery and personalized messaging; tripled throughput via async FastAPI endpoints.

SELECTED PROJECTS

VoxForge — Voice AI Platform Live · 354+ tests

Python · FastAPI · LangGraph · LiveKit · pgvector · Docker · OpenTelemetry

- Built and shipped an open-source voice AI platform solo, live at **voxforge.brohammad.tech** — one pipeline unifying WebSocket, REST, and LiveKit WebRTC with LangGraph agents and pgvector RAG.
- Hardened it with **354+ automated tests (81% coverage)**, Playwright browser tests, per-turn evaluation, signed replay, human handoff, OpenTelemetry, and one-command HTTPS deploy.

BugSathi — Bug Reporting Platform 116 tests · 34 ADRs

Go · Kafka · PostgreSQL · MinIO · Redis · ffmpeg · OpenTelemetry · Kubernetes

- Built a local-first bug reporting platform in Go — an ordered **8-topic Kafka pipeline** from MinIO upload through ffmpeg frame extraction to AI-drafted reports and shareable links.
- Shipped **28 REST endpoints**, 10 Postgres migrations, JWT refresh rotation, and project tenancy; 116 tests run under **-race** in CI alongside OTel/Prometheus/Grafana, Docker/K8s, and 34 ADRs.

SignalMine — Market Intelligence Pipeline 124 tests · 4 JSON schemas

Python · FastAPI · PostgreSQL · Pydantic · Instructor · OpenTelemetry · Prometheus

- Built a source-agnostic pipeline that turns GitHub and Reddit threads into **schema-validated, evidence-backed** complaint records — collectors, a deterministic normalizer, an LLM extractor, and a JSON Schema validator behind one contract.
- Shipped a FastAPI service with Postgres persistence, OpenTelemetry traces, and Prometheus metrics, backed by **124 tests** and a golden-dataset harness that scores extraction field by field.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, C++

Backend: FastAPI, Node.js, Express, REST APIs, gRPC, AsyncIO, WebSockets, SSE, Redis, Kafka

AI / ML: LangGraph, LangChain, RAG, MCP, multi-agent systems, LLM evaluation, prompt engineering, NLP, OpenAI, Gemini

Data, Infra & Testing: PostgreSQL, SQLite, Neo4j, ChromaDB, Qdrant, Docker, Kubernetes, GitHub Actions, Prometheus, Grafana, OpenTelemetry, AWS, GCP, Linux, Git, pytest, Playwright

EDUCATION

National Institute of Technology Calicut B.Tech in Electrical & Electronics Engineering · 2021 – 2025

LEADERSHIP

- **Joint Secretary, Enquire (NIT Calicut):** Drove **20,000+ students** statewide for Samasya, Kerala's largest science competition.
- **IEEE NIT Calicut:** 10,000+ participants across 40+ events; named **Best Student Branch in India**.